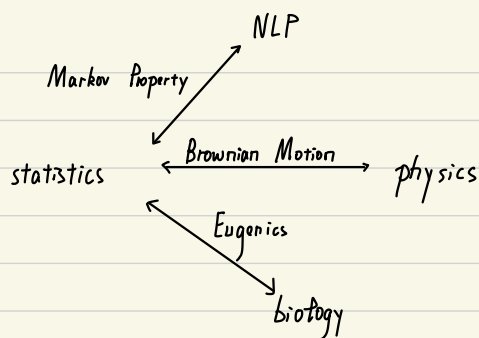
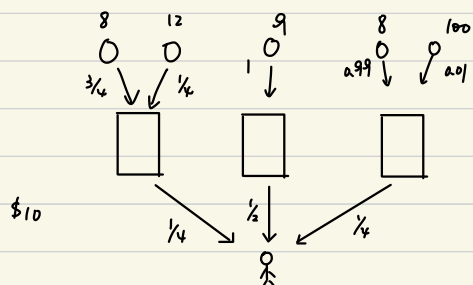


History



RL example



$$E(\text{money}) = E_{\text{machine}}(E_{\text{money}}(\text{money} | \text{machine}))$$

$$\text{Var}(\text{money}) = E_{\text{machine}}(\text{Var}_{\text{money}}(\text{money} | \text{machine})) + \text{Var}_{\text{machine}}(E_{\text{money}}(\text{money} | \text{machine}))$$

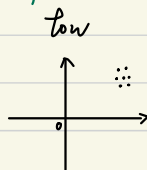
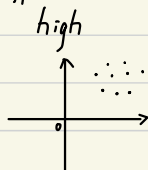
Distribution of money: easy

Bias - Variance - Tradeoff

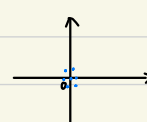
// omit if no time

Bias \ Variance

high



low



$$12D, \vec{y}_{\text{target}} = (0, 0)$$

Use least square as an example

suppose we have



we can define

$$\text{MSE}(y, \hat{y}) \triangleq E[(y - \hat{y})^2]$$

$$\text{Bias}(y, \hat{y}) \triangleq y - E[\hat{y}]$$

$$\text{Var}(\hat{y}) \triangleq E[(\hat{y} - E[\hat{y}])^2]$$

// 这么定义 MSE 出于和定义 Variance 同样的考虑
// 若 $E[\hat{y}] = y$, 则称 $\hat{y}$ 是 y 的无偏估计 (unbiased estimation)

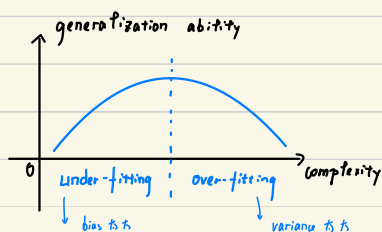
r.v. or number?

sample 后, 你可以这样去把 MSE 算出来, MSE 是一个数
sample 前, MSE is r.v.

$$\text{MSE} = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

y 是一个只有上帝知道的 Const, $\text{Var}(y) = 0$

core of ML



CLT

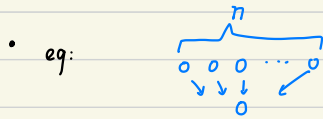
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

$$\cdot X_1, \dots, X_n \text{ iid} \Rightarrow \sum_{i=1}^n X_i \xrightarrow{d} N(nE(X_i), n\text{Var}(X_i))$$

• eg: $\sum X_i \sim B(1, p)$ 时

$$\cdot \sum_{i=1}^n X_i \xrightarrow{d} N(np, np(1-p)) \quad // \text{Gallon board random walk}$$

$$\cdot \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{d} N(p, \frac{1}{n} p(1-p)) \quad // \text{imagine you will get \$1 or \$0 in each } B(1, p)$$



$$\cdot y = \sum_{i=1}^n w_i x_{ni} = \sum w_i x_i \quad w \text{ is weight, you can write } y = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\cdot X_1, \dots, X_n \text{ iid} \Rightarrow y \xrightarrow{d} N(E(X_i), \frac{1}{n} \text{Var}(X_i)) = N(E(X_i), 0) \quad \text{which contradicts exp results}$$

• Z resolution // I think no $\hat{\Sigma}$ if, or they will win a Nobel Prize

$$\cdot E(Y) \approx 0$$

• $X_1, \dots, X_n$ are not independent

• Lady Tasting Tea

$$\cdot P(\text{data} | H_0) = 5.47\% \Leftrightarrow \begin{cases} P(8 | H_0) = C_{10}^8 (1/2)^{10} \approx 4.39\% \\ P(9 | H_0) = C_{10}^9 (1/2)^{10} \approx 0.98\% \\ P(10 | H_0) = C_{10}^{10} (1/2)^{10} \approx 0.10\% \end{cases}$$

• 2 side vs 1 side

• Bayes

$$\cdot P(H_0 | \text{data}) = \frac{P(\text{data} | H_0) P(H_0)}{P(\text{data})}$$

$$\cdot \begin{cases} H_0: \text{She will Agree} \\ H_1: \text{She will NOT Agree} \end{cases} \Rightarrow \frac{P(H_0 | \text{data})}{P(H_1 | \text{data})} = \frac{P(\text{data} | H_0)}{P(\text{data} | H_1)} \frac{P(H_0)}{P(H_1)} \Rightarrow \frac{P(H_0)}{P(H_1)} \leftarrow \frac{P(\text{data} | H_0)}{P(\text{data} | H_1)} \frac{P(H_0)}{P(H_1)}$$

• data: chat, run, eat, movie, poem, music, study ...

• You are using it, even if you don't know

• Sadly, $p(H_0)/p(H_1)$ can not be $+\infty$

• More sophisticated model

• add H_2

$$\cdot \begin{cases} H_0: \text{Finally, She will Agree} \\ H_1: \text{Finally, She will NOT Agree} \end{cases}$$

- 1000 - door - problem

- notion: A is the chosen door, B is the other door

- H_A : A has the goat
 - H_B : B has the goat

- $$\begin{aligned} P(H_A | \text{data}) &= \frac{P(\text{data} | H_A) P(H_A)}{P(\text{data})} = \frac{\frac{1}{999} \cdot \frac{1}{1000}}{\frac{1}{999}} = \frac{1}{1000} \\ P(H_B | \text{data}) &= \frac{P(\text{data} | H_B) P(H_B)}{P(\text{data})} = \frac{1 \cdot \frac{1}{1000}}{\frac{1}{999}} = \frac{999}{1000} \end{aligned}$$

- or you can use
$$\frac{P(H_A | \text{data})}{P(H_B | \text{data})} = \frac{P(\text{data} | H_A)}{P(\text{data} | H_B)} \cdot \frac{P(H_A)}{P(H_B)} = \frac{1}{999}$$

- 1000 - choice - problem

- notion: A is the chosen choice, B is the other choice

- H_A : A is right
 - H_B : B is right

- $$\begin{aligned} P(H_A | \text{data}) &= \frac{P(\text{data} | H_A) P(H_A)}{P(\text{data})} = \frac{\frac{1}{999} \cdot \frac{1}{1000}}{\frac{1}{C_{1000}^2}} = \frac{1}{2} \\ P(H_B | \text{data}) &= \frac{P(\text{data} | H_B) P(H_B)}{P(\text{data})} = \frac{\frac{1}{999} \cdot \frac{1}{1000}}{\frac{1}{C_{1000}^2}} = \frac{1}{2} \end{aligned}$$

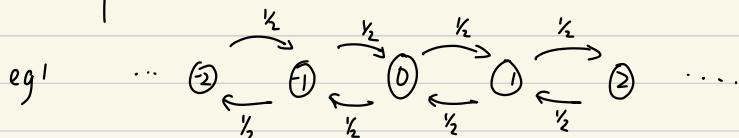
- or you can use
$$\frac{P(H_A | \text{data})}{P(H_B | \text{data})} = \frac{P(\text{data} | H_A)}{P(\text{data} | H_B)} \cdot \frac{P(H_A)}{P(H_B)} = 1$$

> Stochastic process

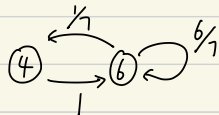
• Markov process:

- Markov Property: 未来只和现在有关, 和过去无关

• 离散: def $P\{\Delta_{n+1} = i_{n+1} \mid \Delta_0 = i_0, \Delta_1 = i_1, \dots, \Delta_{n-1} = i_{n-1} \mid \Delta_n = i_n\} = P\{\Delta_{n+1} = i_{n+1} \mid \Delta_n = i_n\}$

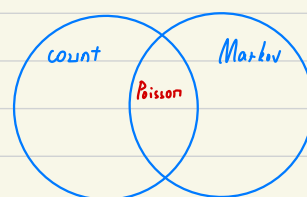


eg2



• 连续: def $P(\Delta(t+s) = j \mid \Delta(s) = i, \Delta(u) = x(u), 0 \leq u \leq s) = P(\Delta(t+s) = j \mid \Delta(s) = i) \forall s, t \forall i, j, x(u) \forall u$

• Poisson process: def $(0) \xrightarrow{1} (1) \xrightarrow{1} (2) \xrightarrow{1} (3) \rightarrow \dots$



• $t_{n+1} - t_n \sim \text{Exp}$

• $N(t_1 + t_2) - N(t_1) \sim \text{Poi}$

• steady

• 过程平稳性

• 若 $\forall t_1, \dots, t_n \forall h (\Delta(t_1+h), \dots, \Delta(t_n+h)) \stackrel{d}{=} (\Delta(t_1), \dots, \Delta(t_n))$ 则称 Δ 有严平稳性

• 若 $E[\Delta(t)] = \text{Const}$ 则称 Δ 有宽平稳性

$\text{Cov}(\Delta(t_1), \Delta(t_2)) = f(t_1 - t_2)$

$E[\Delta(t)^2] < +\infty$

• 遍历性: $\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (1 - \frac{t}{T}) R_2(t) dt = 0 \Leftrightarrow \Delta(t, \omega)$ 有遍历性 $\Leftrightarrow \langle \Delta(t, \omega) \rangle_\omega = \langle \Delta(t, \omega) \rangle_t$

• 过程增量平稳性: 若 $\forall t_1, t_2 \Delta(t_1+h) - \Delta(t_1) \stackrel{d}{=} \Delta(t_2+h) - \Delta(t_2)$ 则称 Δ 的增量有平稳性

• 过程增量的独立性: 若 $\forall t_1 < t_2 < \dots < t_n \Delta(t_2) - \Delta(t_1), \Delta(t_3) - \Delta(t_2), \dots, \Delta(t_n) - \Delta(t_{n-1})$ 独立 则称 Δ 的增量有独立性

> In Neuroscience

• $p(t) = \sum_{i=1}^n \delta(t - t_i)$ $\psi(p(t, z))$ $f(t, z)$

$\langle f(t, z) \rangle_t \xrightarrow[\text{time average}]{\frac{1}{T} \int_0^T \rho(t) dt} \xrightarrow[\text{trial average}]{\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \frac{1}{T} \int_0^T \rho_i(t) dt} \xrightarrow[\text{time average}]{\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \langle \rho_i(t) \rangle_t} \langle f(t, z) \rangle_\omega$

$\langle f(t, z) \rangle_\omega$ usually means $\langle f(t, z) \rangle_\omega$

• $C(z) \triangleq \begin{cases} \langle \frac{1}{n} \sum_{i=1}^n s(t_i - z) \rangle & t \in [0, t_{\text{threshold}}] \\ 0 & z > t_{\text{threshold}} \\ 0 & z < 0 \end{cases}$

• When n is large and trial-trial variability is not too much, $C(z) \approx \frac{1}{n} \langle \sum_{i=1}^n s(t_i - z) \rangle$

• $C(z) = \frac{1}{\langle n \rangle} \int_0^{t_{\text{exp}}} dt \langle \rho(t) \rangle s(t - z) \stackrel{\text{Dirac function}}{=} \frac{1}{\langle n \rangle} \int_0^{t_{\text{exp}}} dz \rho(z) s(t - z) = \frac{1}{\langle n \rangle} R_{s, \rho}(z) \quad z \in [0, t_{\text{threshold}}]$

$R_{s, \rho}(z) \triangleq \frac{1}{t_{\text{exp}}} \int_0^{t_{\text{exp}}} s(t - z) \rho(t) dt$